DSP - SECTION FIVE

System Properties

Proving linearity, time invariance & causality with solved problems

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TODAY'S PLAN

Three questions we ask of every system

01

Linear?

Does superposition hold? The output of a weighted sum must equal the weighted sum of the outputs.

02

Time-invariant?

If the input is delayed by n_0, is the output delayed by exactly n_0 and nothing else?

03

Causal?

Does y(n) use only present and past inputs, never future ones like x(n+1)?

PROPERTY 1 · LINEARITY

Linear means superposition holds

DEFINITION

If $x_1(n) \rightarrow y_1(n)$
and $x_2(n) \rightarrow y_2(n)$
then
$\alpha x_1(n) + \beta x_2(n) \rightarrow \alpha y_1(n) + \beta y_2(n)$

α and β are any constants.

How to prove it, in 3 steps

1. Find each output alone: $y_1 = T\{x_1\}$, $y_2 = T\{x_2\}$
2. Feed the combined input: $y_3 = T\{\alpha x_1 + \beta x_2\}$
3. Compare y_3 with $\alpha y_1 + \beta y_2$. Equal → linear. Not equal → nonlinear.

Quick check: zero in must give zero out. A constant term (+3) or a power (x^2) breaks linearity.

LINEARITY · EXAMPLE 1 (LECTURE)

Is the amplifier y(n) = 10x(n) linear?

Given: $x_1(n) = u(n)$, $x_2(n) = \delta(n)$, $\alpha = 2$, $\beta = 4$

1 · Each alone

$$y_1(n) = 10u(n) \qquad y_2(n) = 10\delta(n)$$

2 · Combined input

$$x(n) = 2u(n) + 4\delta(n)$$
$$y(n) = 10[2u(n) + 4\delta(n)] = 20u(n) + 40\delta(n)$$

3 · Compare

$$2y_1(n) + 4y_2(n) = 20u(n) + 40\delta(n)$$

Linear Step 2 = Step 3, so $y(n) = 2y_1(n) + 4y_2(n)$. Superposition holds.

LINEARITY · EXAMPLE 2 (LECTURE)

Is the squarer y(n) = x²(n) linear?

Given: $x_1 = u(n)$, $x_2 = \delta(n)$, $\alpha = 4$, $\beta = 2$. Facts: $u^2 = u$, $\delta^2 = \delta$, $u \cdot \delta = \delta$

1 · Each alone

$$y_1(n) = u^2(n) = u(n) \qquad y_2(n) = \delta^2(n) = \delta(n)$$

2 · Combined input

$$\begin{aligned} y(n) &= (4u(n) + 2\delta(n))^2 \\ &= 16u^2 + 16u\cdot\delta + 4\delta^2 \\ &= 16u + 16\delta + 4\delta = 16u(n) + 20\delta(n) \end{aligned}$$

3 · Compare

$$4y_1(n) + 2y_2(n) = 4u(n) + 2\delta(n)$$

Nonlinear 16u(n) + 20δ(n) ≠ 4u(n) + 2δ(n). Superposition fails.

LINEARITY · EXAMPLE 3 (NUMERICAL PROOF)

Check y[n] = 2x[n] with real numbers

Given: $x_1[n] = \{1, 2, 3\}$, $x_2[n] = \{4, 5, 6\}$, $a = 2$, $b = 3$

Step	n = 0	n = 1	n = 2
1 · $y_1 = 2x_1$	2	4	6
1 · $y_2 = 2x_2$	8	10	12
2 · $ax_1 + bx_2$	14	19	24
2 · $T\{ax_1 + bx_2\} = 2(\ldots)$	28	38	48
3 · $ay_1 + by_2$	4+24 = 28	8+30 = 38	12+36 = 48

Linear The two shaded rows match at every n: {28, 38, 48} = {28, 38, 48}.

LINEARITY · EXAMPLE 4 (NUMERICAL PROOF)

Check $y[n] = x^2[n]$ with the same numbers

Given: $x_1[n] = \{1, 2, 3\}$, $x_2[n] = \{4, 5, 6\}$, $a = 2$, $b = 3$

Step	n = 0	n = 1	n = 2
1 · $y_1 = x_1^2$	1	4	9
1 · $y_2 = x_2^2$	16	25	36
2 · $ax_1 + bx_2$	14	19	24
2 · $T\{ax_1 + bx_2\} = (\ldots)^2$	196	361	576
3 · $ay_1 + by_2$	2+48 = 50	8+75 = 83	18+108 = 126

Nonlinear {196, 361, 576} ≠ {50, 83, 126}. One mismatch is enough.

LINEARITY · PRACTICE 1

Is y(n) = 2x(n) + 3 linear?

Given: any inputs $x_1(n)$, $x_2(n)$ and any constants α, β

1 · Each alone

$$y_1(n) = 2x_1(n) + 3 \qquad y_2(n) = 2x_2(n) + 3$$

2 · Combined input

$$\begin{aligned} T\{\alpha x_1 + \beta x_2\} &= 2[\alpha x_1(n) + \beta x_2(n)] + 3 \\ &= 2\alpha x_1(n) + 2\beta x_2(n) + 3 \end{aligned}$$

3 · Compare

$$\alpha y_1(n) + \beta y_2(n) = 2\alpha x_1(n) + 2\beta x_2(n) + 3\alpha + 3\beta$$

Nonlinear The constant gives 3 in step 2 but 3(α + β) in step 3. Also x = 0 gives y = 3, not 0.

LINEARITY · PRACTICE 2

Is y(n) = n·x(n) linear?

Given: any inputs $x_1(n)$, $x_2(n)$ and any constants α, β

1 · Each alone

$$y_1(n) = n \cdot x_1(n) \qquad y_2(n) = n \cdot x_2(n)$$

2 · Combined input

$$T\{\alpha x_1 + \beta x_2\} = n[\alpha x_1(n) + \beta x_2(n)] = \alpha n \cdot x_1(n) + \beta n \cdot x_2(n)$$

3 · Compare

$$\alpha y_1(n) + \beta y_2(n) = \alpha n \cdot x_1(n) + \beta n \cdot x_2(n)$$

Linear Multiplying by n (a function of time, not of x) keeps superposition. Remember this one for time invariance.

PROPERTY 2 · TIME INVARIANCE

Shift the input, the output shifts the same way

DEFINITION

If $x_1(n) \rightarrow y_1(n)$

let $x_2(n) = x_1(n - n_0)$

then the system is TI if

$y_2(n) = y_1(n - n_0)$

n_0 is any whole-number delay. The system's rule must not change with time.

How to prove it, in 3 steps

1. **Shift the input:** $y_2(n) = T\{x_1(n - n_0)\}$. Replace every $x(\cdot)$ with $x_1(\cdot - n_0)$.

2. **Shift the output:** $y_1(n - n_0)$. Replace EVERY n with $(n - n_0)$, inside and outside x.

3. Equal → time-invariant. Not equal → time-variant.

Red flags: n outside x, like $n \cdot x(n)$; scaled time $x(3n)$; reversed time $x(-n)$.

TIME INVARIANCE · EXAMPLE 1 (LECTURE)

Is y(n) = 2x(n – 5) time-invariant?

Start: $y_1(n) = 2x_1(n - 5)$, shifted input $x_2(n) = x_1(n - n_0)$

1 · Shift input	$y_2(n) = 2x_2(n - 5) = 2x_1(n - 5 - n_0)$
2 · Shift output	$y_1(n - n_0) = 2x_1((n - n_0) - 5) = 2x_1(n - 5 - n_0)$
3 · Compare	$y_2(n) = y_1(n - n_0)$

Time-invariant Delaying the input by n_0 delays the output by the same n_0.

TIME INVARIANCE · EXAMPLE 2 (LECTURE)

Is y(n) = 2x(3n) time-invariant?

Start: $y_1(n) = 2x_1(3n)$, shifted input $x_2(n) = x_1(n - n_0)$

1 · Shift input	$y_2(n) = 2x_2(3n) = 2x_1(3n - n_0)$
2 · Shift output	$y_1(n - n_0) = 2x_1(3(n - n_0)) = 2x_1(3n - 3n_0)$
3 · Compare	$3n - n_0 \neq 3n - 3n_0 \rightarrow y_2(n) \neq y_1(n - n_0)$

Time-variant Compressing time (3n) turns an input shift of n_0 into an output shift of $n_0/3$.

TIME INVARIANCE · PRACTICE 1

Is y(n) = n·x(n) time-invariant?

Start: $y_1(n) = n \cdot x_1(n)$, shifted input $x_2(n) = x_1(n - n_0)$

1 · Shift input	$y_2(n) = n \cdot x_2(n) = n \cdot x_1(n - n_0)$
2 · Shift output	$y_1(n - n_0) = (n - n_0) \cdot x_1(n - n_0)$
3 · Compare	$n \neq (n - n_0) \rightarrow y_2(n) \neq y_1(n - n_0)$

Time-variant The gain n changes with time. So this system is linear but time-variant.

TIME INVARIANCE · PRACTICE 2

Two quick proofs side by side

(a) $y(n) = x^2(n)$

1 · Shift input

$$y_2(n) = x_2^2(n) = x_1^2(n - n_0)$$

2 · Shift output

$$y_1(n - n_0) = x_1^2(n - n_0)$$

3 · Compare

equal

Time-invariant (but nonlinear)

(b) $y(n) = x(-n)$

1 · Shift input

$$y_2(n) = x_2(-n) = x_1(-n - n_0)$$

2 · Shift output

$$y_1(n - n_0) = x_1(-(n - n_0)) = x_1(-n + n_0)$$

3 · Compare

$$-n - n_0 \neq -n + n_0$$

Time-variant

PROPERTY 3 · CAUSALITY

Causal means no peeking at the future

Causal

y(n) uses only the present input x(n) and past inputs x(n − 1), x(n − 2), … (past outputs y(n − 1) are fine too).

Noncausal

y(n) needs at least one future input such as x(n + 1), x(n + 2), … It cannot be built in real time.

HOW TO CHECK

Look at every x(·). Compare its argument with n. If every argument is ≤ n → causal. If any argument can be > n → noncausal. Stuck? Plug in a value, e.g. n = 1, and list the inputs you need.

CAUSALITY · EXAMPLE 3.4 (LECTURE)

Read the arguments, decide

(a) $y(n) = 0.5x(n) + 2.5x(n - 2), \quad n \geq 0$

Inputs used: $x(n)$ → present, $x(n - 2)$ → past

Causal

(b) $y(n) = 0.25x(n - 1) + 0.5x(n + 1) - 0.4y(n - 1), \quad n \geq 0$

Inputs used: $x(n - 1)$ → past, $x(n + 1)$ → FUTURE, $y(n - 1)$ → past output

Noncausal, because of $x(n + 1)$

Note: the lecture's solution for (b) mentions $x(n)$ and $x(n + 2)$; the equation actually contains $x(n - 1)$ and $x(n + 1)$. The answer, noncausal, is the same.

CAUSALITY · PRACTICE

Six systems, one quick test each

System	Sample check	Verdict
$y(n) = x(n) - x(n-1)$	$n = 1$ needs $x(1)$, $x(0)$	Causal
$y(n) = x(n + 1)$	$n = 1$ needs $x(2)$: future	Noncausal
$y(n) = x(2n)$	$n = 1$ needs $x(2)$: future	Noncausal
$y(n) = x(-n)$	$n = -1$ needs $x(1)$: future	Noncausal
$y(n) = \sum_{k=-\infty}^{n} x(k)$	$n = 1$ needs $x(1)$, $x(0)$, $x(-1)$, …	Causal
$y(n) = n \cdot x(n)$	$n = 1$ needs $x(1)$ only	Causal

Watch out: x(2n) and x(−n) look harmless, but for some n their argument is ahead of n.

SUMMARY

Every system we proved today

System	Linear?	Time-inv.?	Causal?
$y(n) = 10x(n)$	Yes	Yes	Yes
$y(n) = x^2(n)$	No	Yes	Yes
$y(n) = 2x(n) + 3$	No	Yes	Yes
$y(n) = n \cdot x(n)$	Yes	No	Yes
$y(n) = 2x(n - 5)$	Yes	Yes	Yes
$y(n) = 2x(3n)$	Yes	No	No
$y(n) = x(-n)$	Yes	No	No
$y(n) = x(n + 1)$	Yes	Yes	No

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Thank You

Any questions?

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